



BLM3620 Digital Signal Processing

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Yıldız Technical University – Computer Engineering

Course Materials



Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.
- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxiliary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.
- Prof. Nizamettin Aydın, DSP Lecture Notes.
- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Pearson, 2014.
- J. K. Perin, *Digital Signal Processing, Lecture Notes*, Stanford University, 2018.

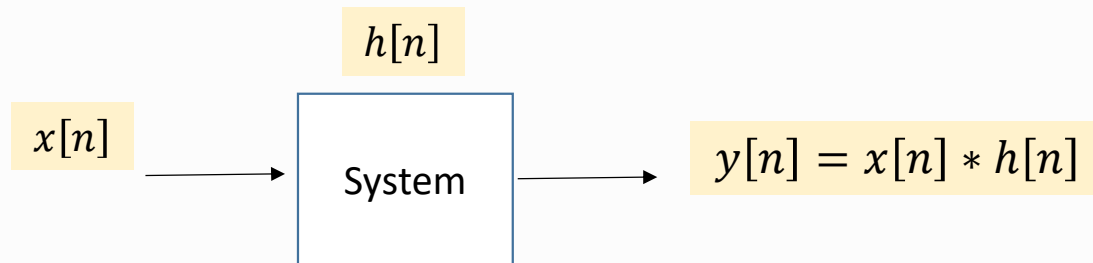
Syllabus



Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	z- Transforms
13	FIR Filter Design and Applications
14	IIR Filter Design and Applications
15	Final Exam

For more details -> Bologna page: <http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3>

Remember: DT Convolution



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

CONVOLUTION SUM

Or

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

There are three approaches to calculate convolution:

- 1) Mathematical Approach
- 2) Table Approach (Polynomial Multiplication)
- 3) Graphical Approach

Lecture #6 – Convolution and FIR Filters

- Convolution Example
- Graphical Convolution
- MATLAB demo
- FIR Filter
- FIR Filter Application

$$x[n] = (0.5)^n u[n]$$

One Example to Mathematical Approach (Study it at home)

Given two signals $a[n] = (0.2)^n u[n]$ ve $b[n] = (0.6)^n u[n]$ find the convolution results $c[n] = a[n] * b[n]$ using mathematical approach.

$$c[n] = \sum_{k=-\infty}^{\infty} a[k]b[n-k]$$

$$c[n] = \sum_{k=-\infty}^{\infty} (0.2)^k u[k] (0.6)^{n-k} u[n-k]$$

$k < 0$ için 0 $k > n$ için 0

$$c[n] = \sum_{k=0}^n (0.2)^k u[k] (0.6)^{n-k} u[n-k]$$

$$c[n] = \sum_{k=0}^n (0.2)^k (0.6)^{n-k}$$

$$\sum_{k=0}^n (0.2)^k (0.6)^{-k} (0.6)^n = (0.6)^n u[n] \sum_{k=0}^n (1/3)^k$$

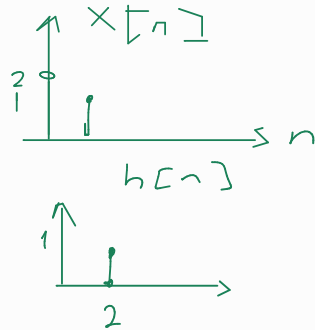
$$c[n] = (0.6)^n u[n] \sum_{k=0}^n \frac{(1/3)^{n+1} - (1/3)^0}{(1/3) - 1}$$

$$c[n] = 2.5(0.6^{n+1} - 0.2^{n+1})u[n]$$

Geometric Serial Sum

$$\sum_{n=M}^N r^k = \frac{r^{N+1} - r^M}{r - 1}$$

Convolution Method – 3: Graphical Approach



For $n=0$,

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]h[-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

For $n=-5$,

$$y[-5] = \sum_{k=-\infty}^{\infty} x[k]h[-5-k]$$



For $n=5$,

$$y[5] = \sum_{k=-\infty}^{\infty} x[k]h[5-k]$$



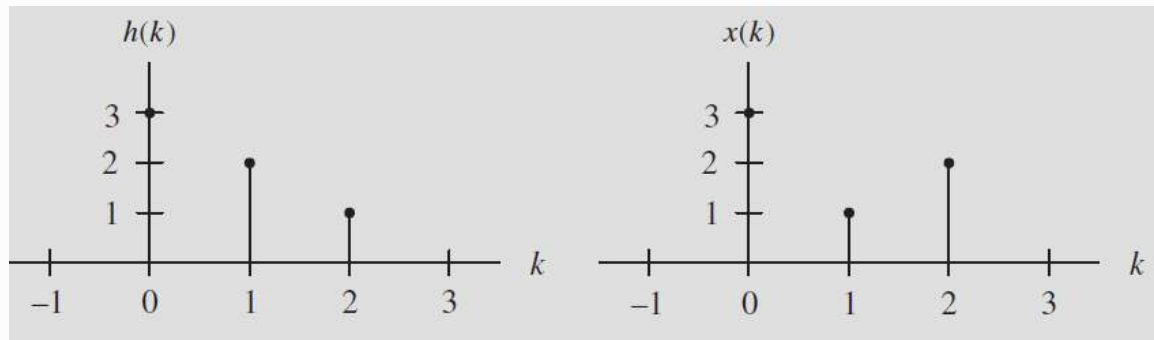
Step 1. Obtain the reversed sequence $h(-k)$.

Step 2. Shift $h(-k)$ by $|n|$ samples to get $h(n-k)$. If $n \geq 0$, $h(-k)$ will be shifted to right by n samples; but if $n < 0$, $h(-k)$ will be shifted to the left by $|n|$ samples.

Step 3. Perform the convolution sum that is the sum of products of two sequences $x(k)$ and $h(n-k)$ to get $y(n)$.

Step 4. Repeat steps (1)–(3) for the next convolution value $y(n)$.

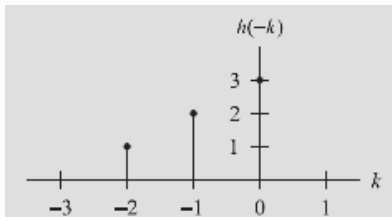
Example for Graphical Approach



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

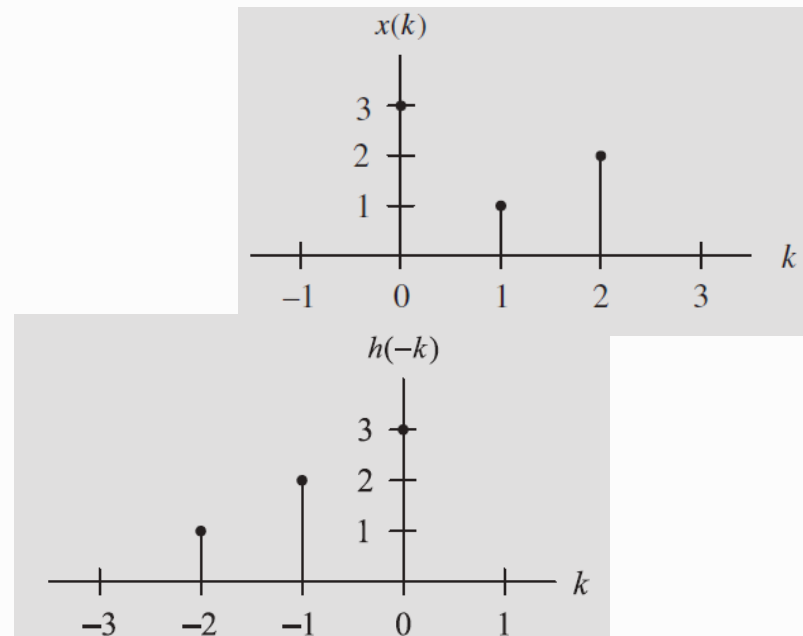
$$y[n] = \sum_{k=0}^2 x[k]h[n-k]$$

1) Obtain $h(-k)$



2) Shift it by 0 and get $h(0-k)$

3) Perform conv. sum

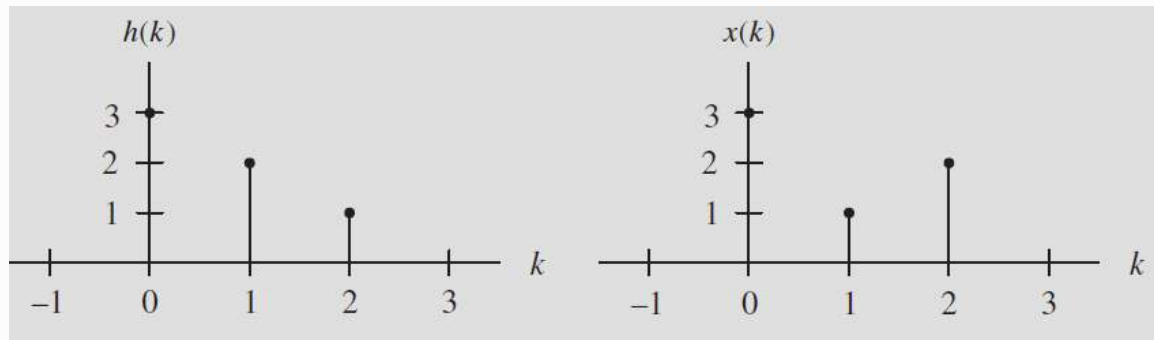


Equal to:

$$y[0] = \sum_{k=0}^2 x[k]h[-k]$$

$$y[0]=9$$

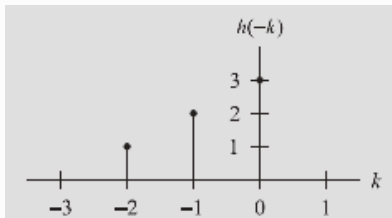
Example for Graphical Approach



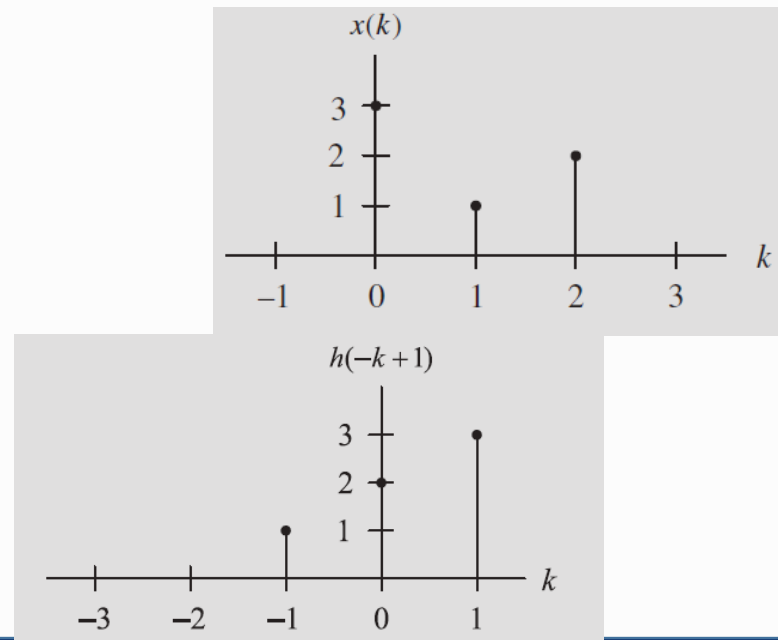
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=0}^3 x[k]h[n-k]$$

1) Obtain $h(-k)$



2) Shift it by 1 and get $h(1-k)$

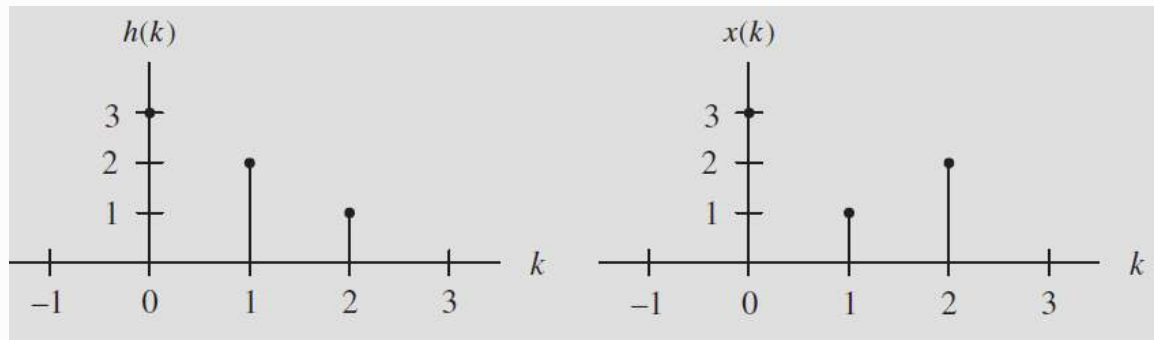


Equal to:

$$y[1] = \sum_{k=0}^3 x[k]h[1-k]$$

$$y[1]=2*3+1*3=9$$

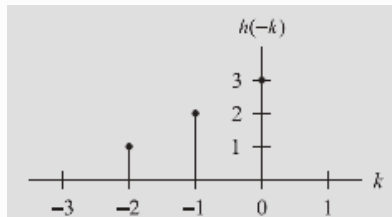
Example for Graphical Approach



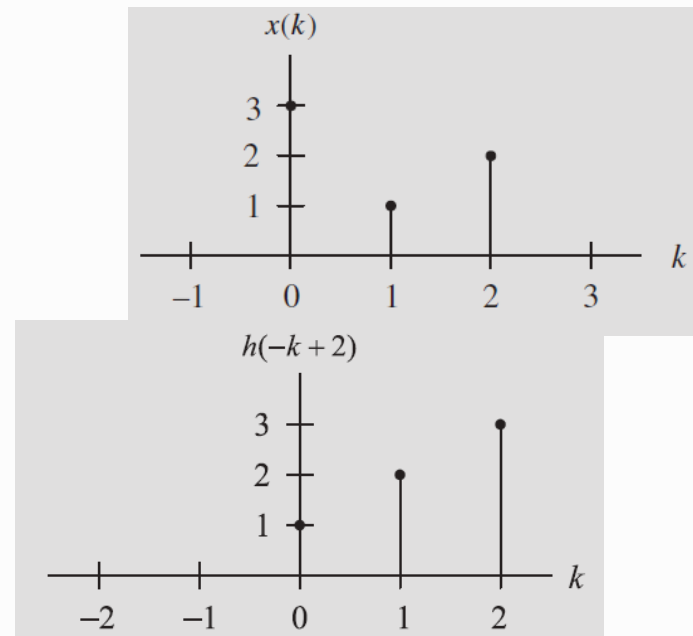
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=0}^3 x[k]h[n-k]$$

1) Obtain $h(-k)$



2) Shift it by 2 and get $h(2-k)$

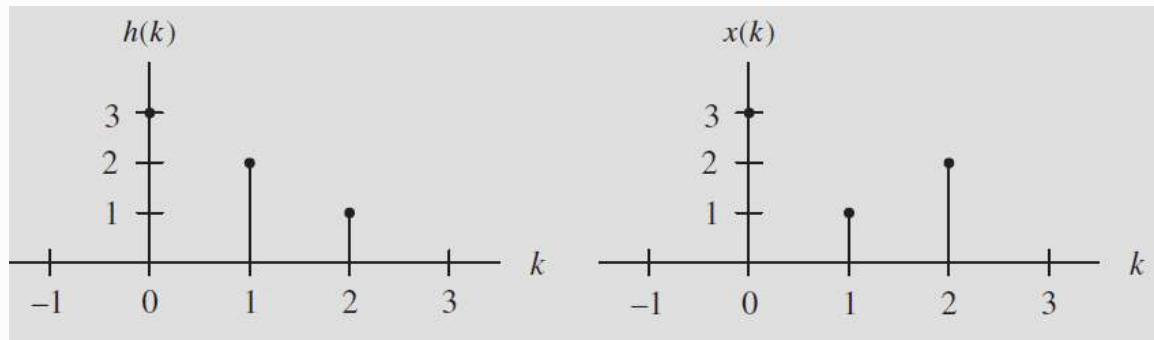


Equal to:

$$y[2] = \sum_{k=0}^3 x[k]h[2-k]$$

$$y[2] = 1*3 + 2*1 + 3*2 = 11$$

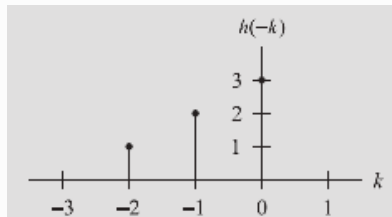
Example for Graphical Approach



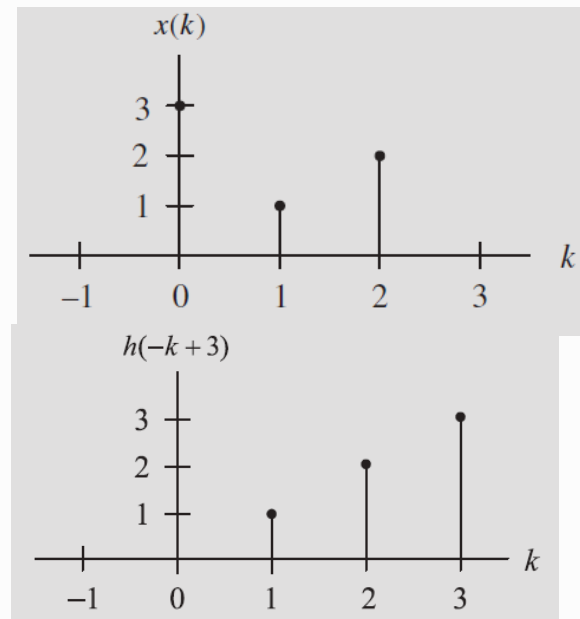
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=0}^3 x[k]h[n-k]$$

1) Obtain $h(-k)$



2) Shift it by 3 and get $h(3-k)$



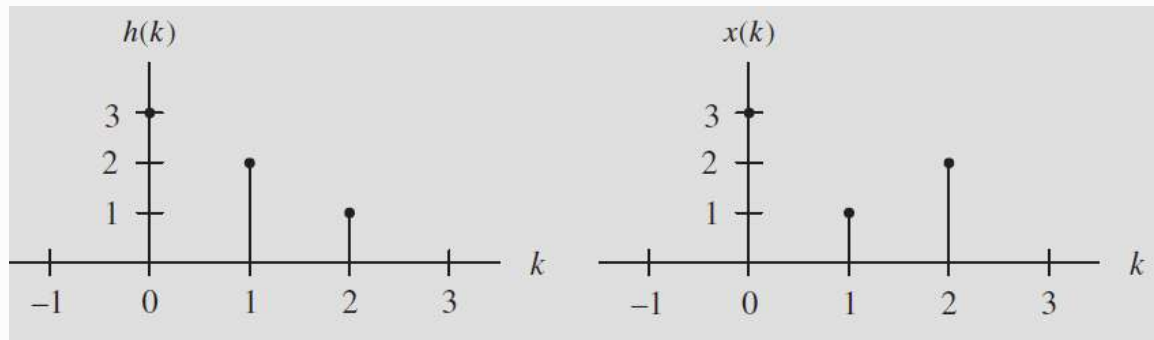
Equal to:

$$y[3] = \sum_{k=0}^3 x[k]h[3-k]$$

3) Perform conv. sum

$$y[3] = 1*1 + 2*2 = 5$$

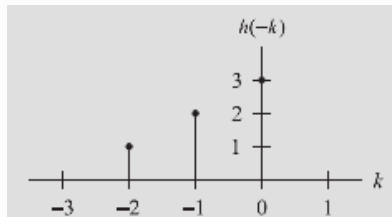
Example for Graphical Approach



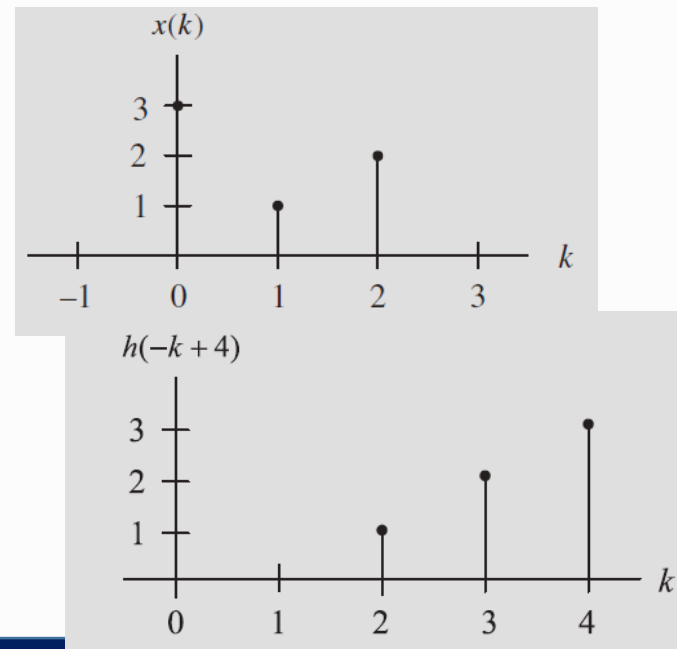
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=0}^3 x[k]h[n-k]$$

1) Obtain $h(-k)$



2) Shift it by 4 and get $h(4-k)$



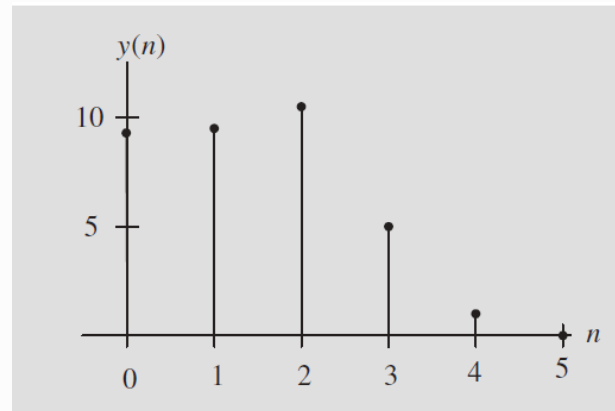
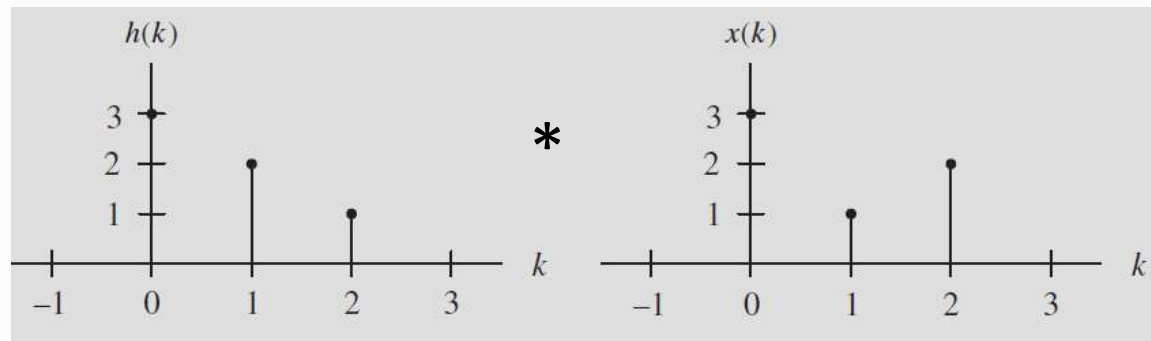
Equal to:

$$y[4] = \sum_{k=0}^3 x[k]h[4-k]$$

$$y[4]=2$$

3) Perform conv. sum

Example for Graphical Approach



$x \rightarrow 312$

$h \rightarrow 321$

$y[0] = 9$

$y[1] = 9$

$y[2] = 11$

3 1 2

3 1 2

3 1 2

1 2 3

1 2 3

1 2 3

$y[3] = 5$

$y[4] = 2$

3 1 2

3 1 2

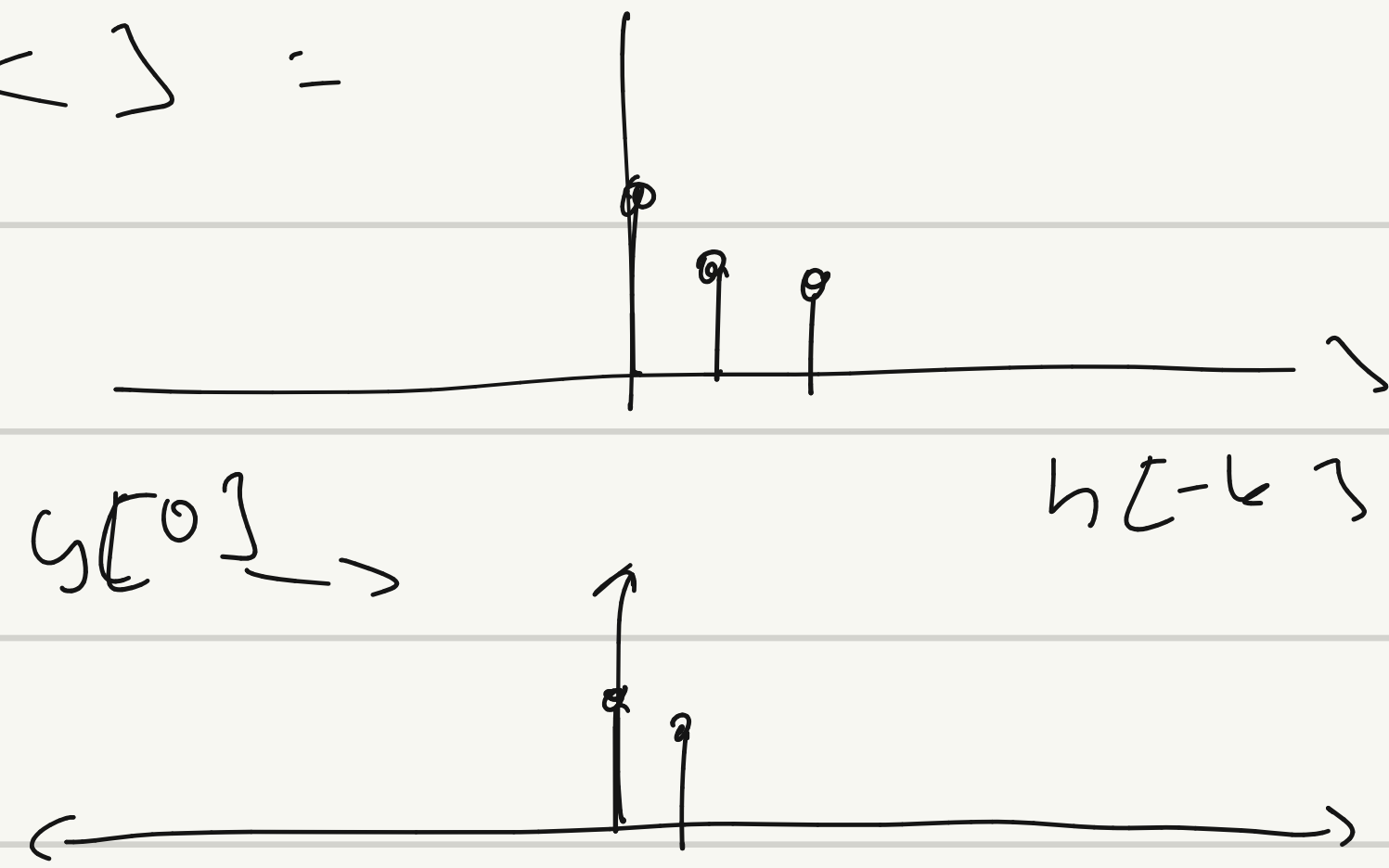
1 2 3

1 2 3

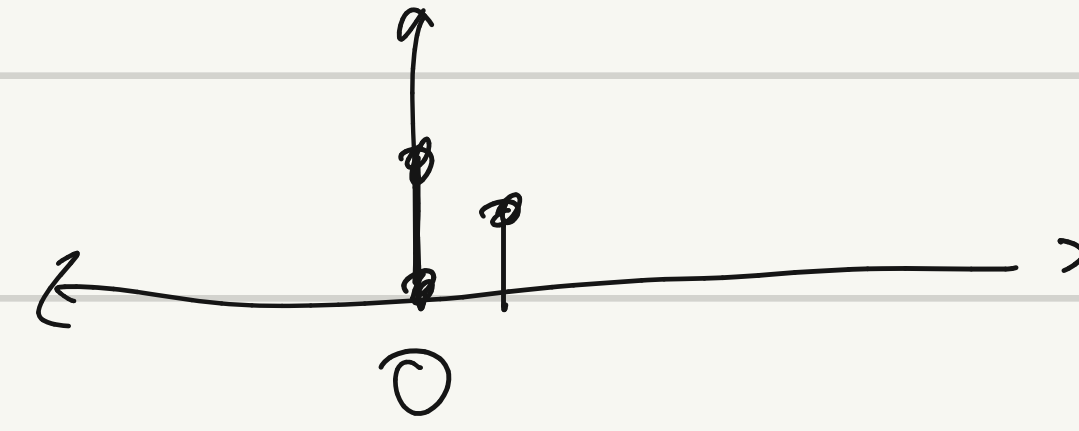
$h[k]$

#

$x[k] =$



$h[-k]$

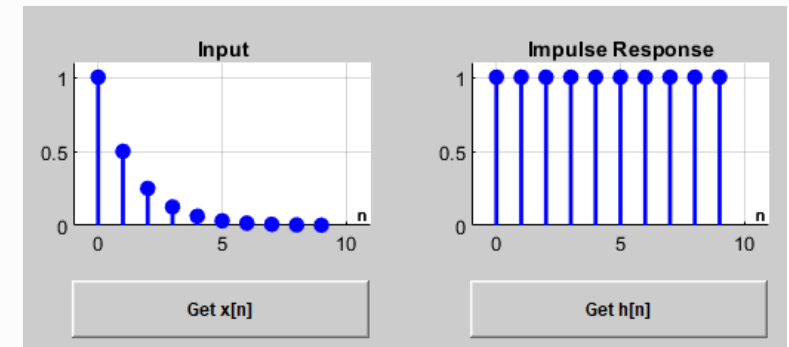
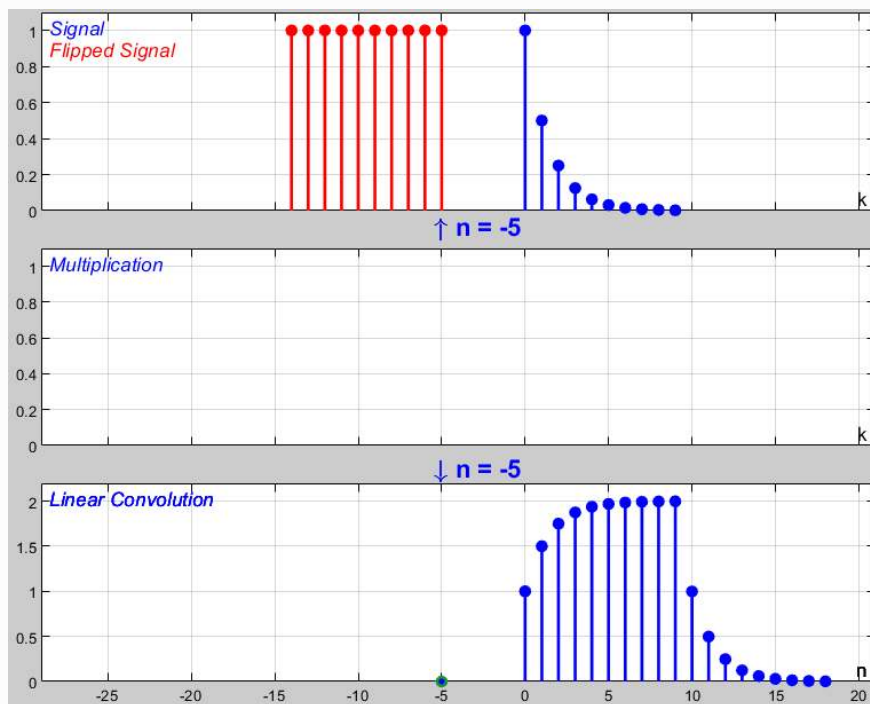


$\rightarrow$ bundan dolayı $y[-1]$ 'den La_1 'a.

MATLAB dconv Demo

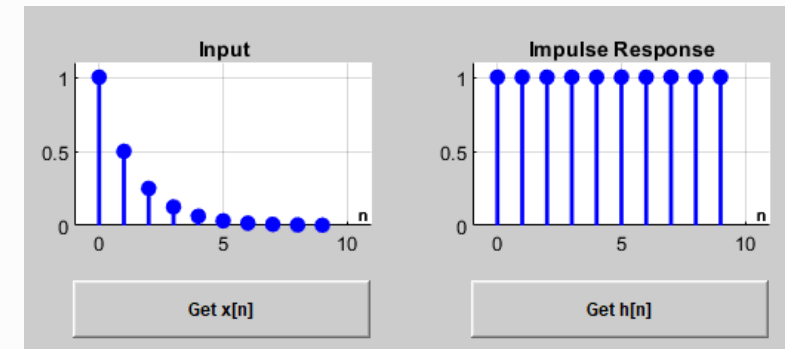
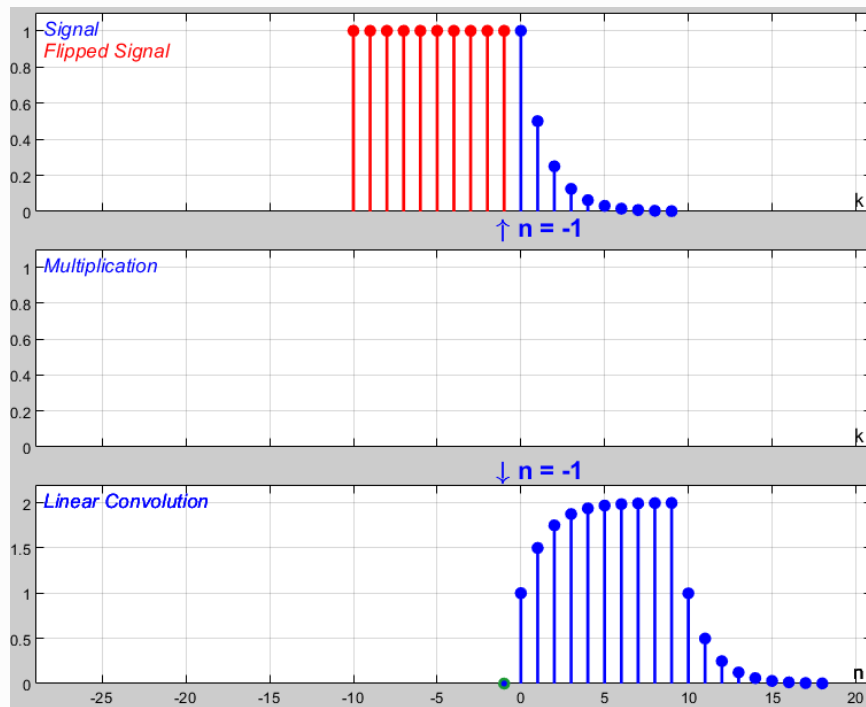


<https://dspfirst.gatech.edu/matlab/#dconvdemo>



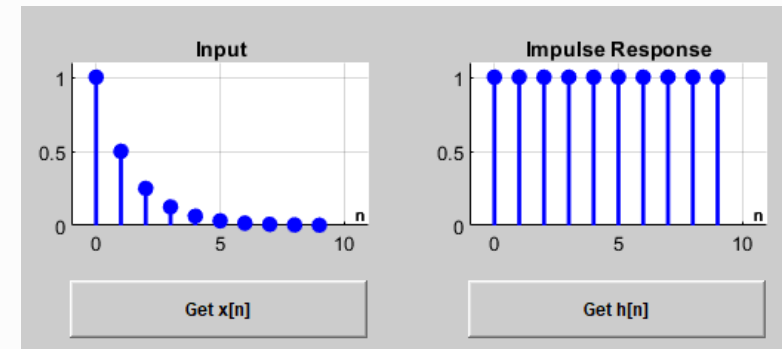
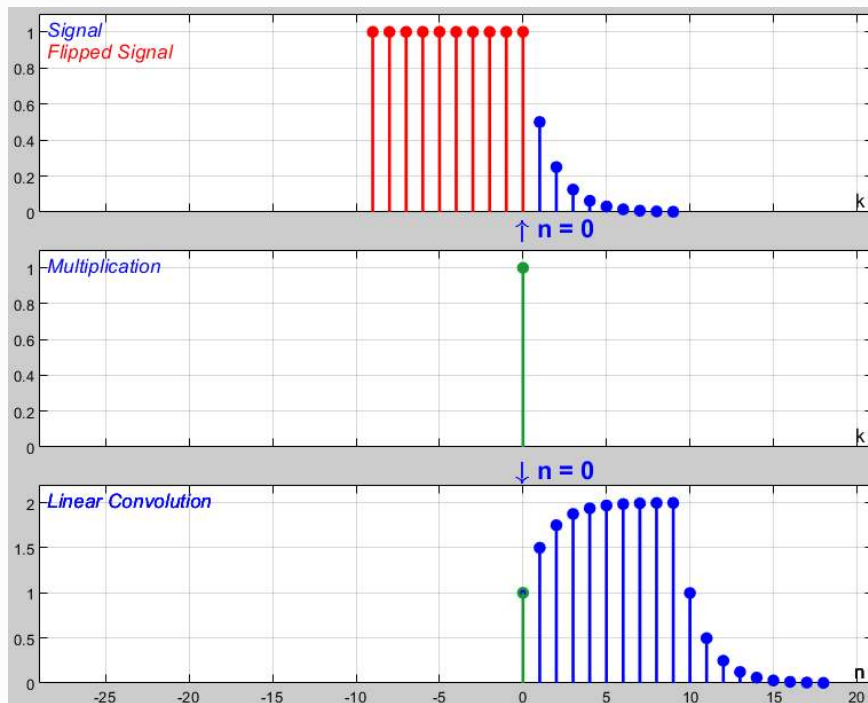
Ref. Discrete Conv. Demo v 3.15 @GitHub

MATLAB dconv Demo



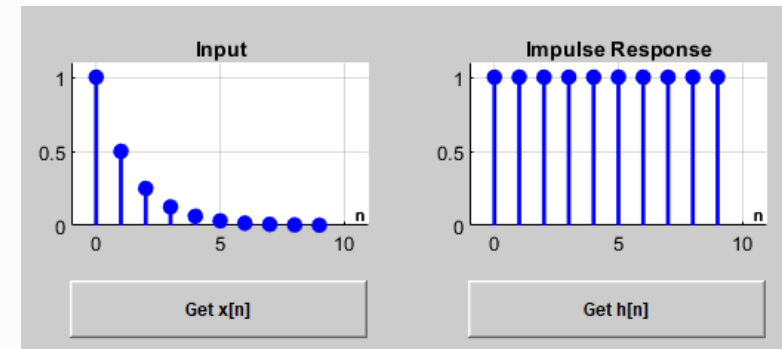
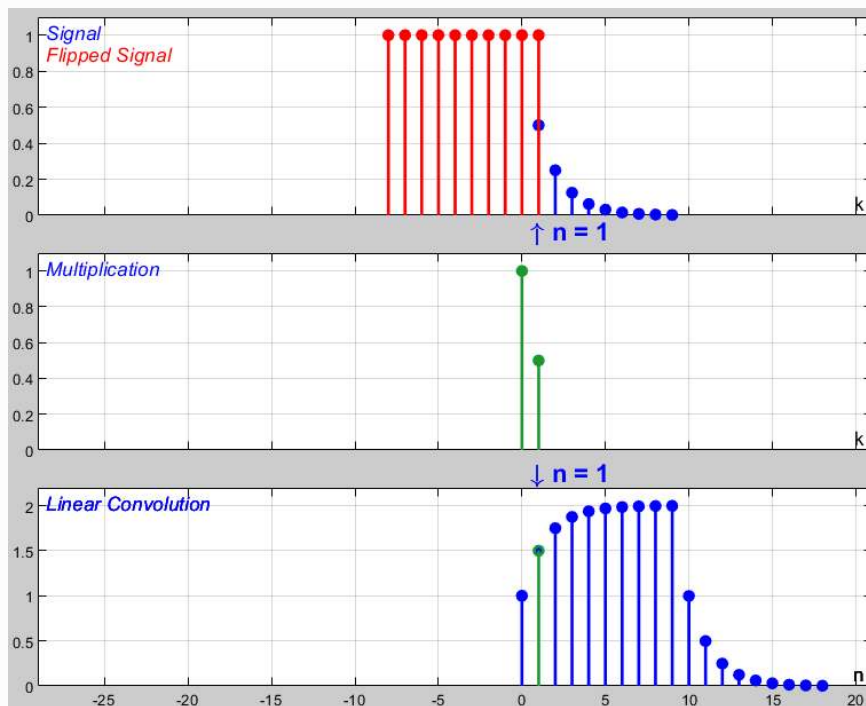
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MATLAB dconv Demo



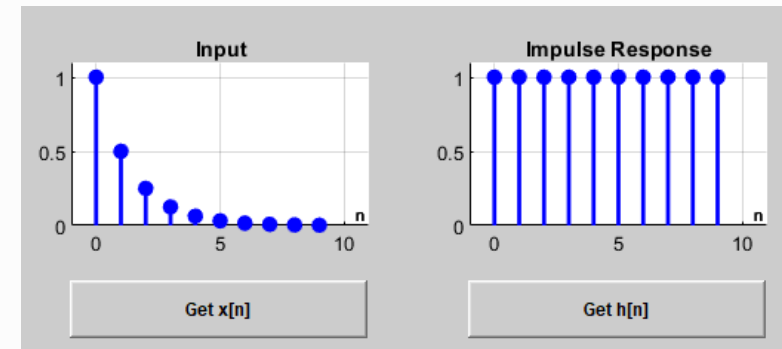
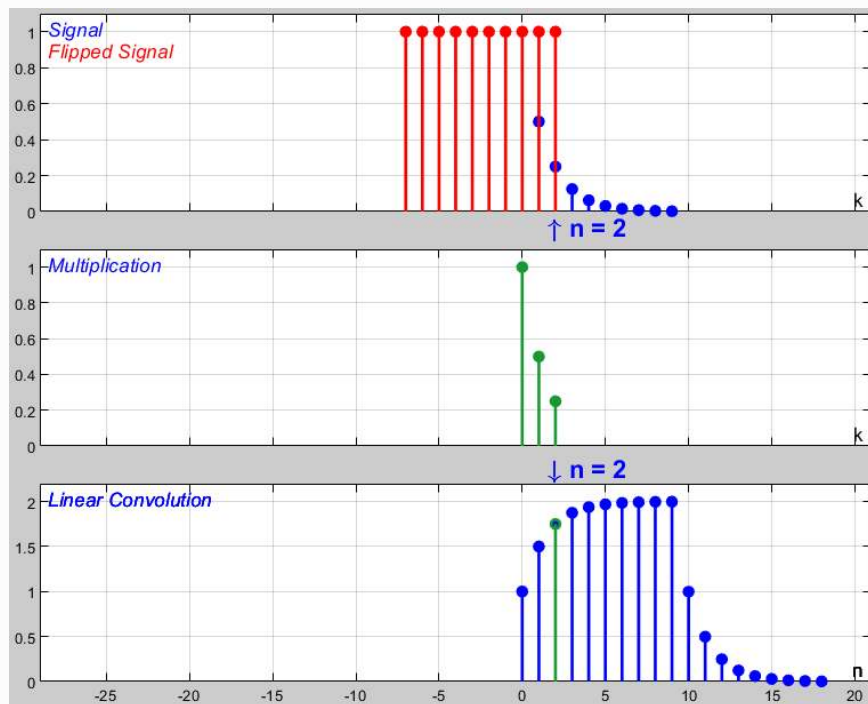
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MATLAB dconv Demo



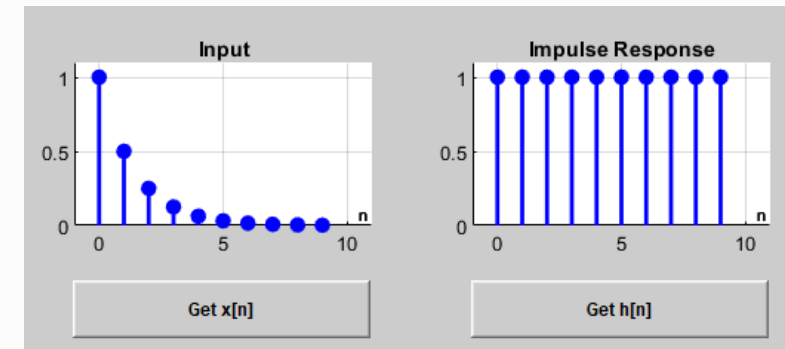
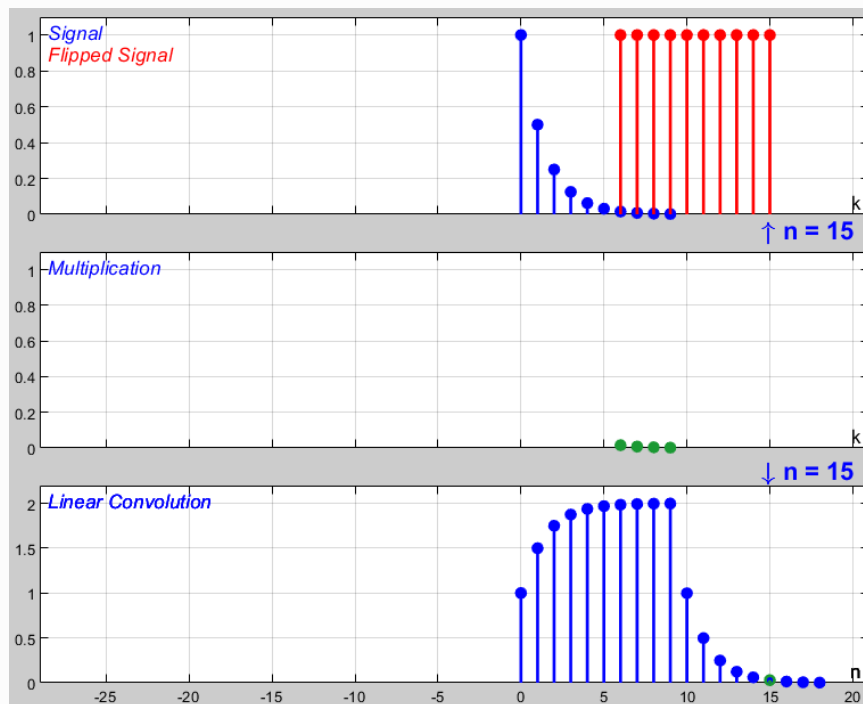
Ref. Discrete Conv. Demo v 3.15 @GitHub

MATLAB dconv Demo



Ref. Discrete Conv. Demo v 3.15 @GitHub

MATLAB dconv Demo



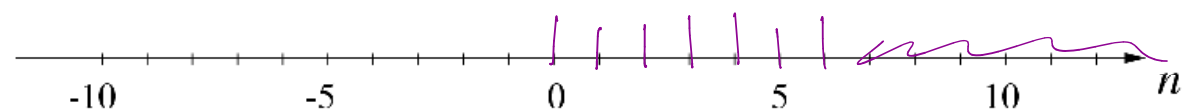
Ref. Discrete Conv. Demo v 3.15 @GitHub

Exercise - 1

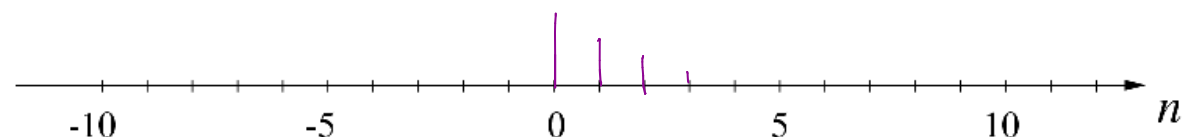
PROBLEM:

Let $x[n] = u[n] - u[n - 7]$ and $h[n] = \begin{cases} (\frac{1}{2})^n & 0 \leq n \leq 3 \\ 0 & \text{otherwise.} \end{cases}$

(a) Plot $x[n]$.



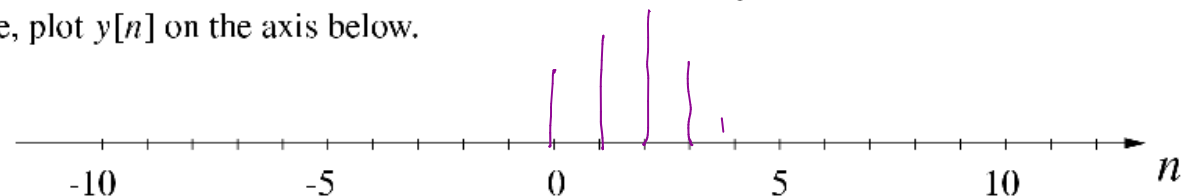
Plot $h[n]$.



Label the amplitudes for each sample.



(b) If we now assume $x[n] = \delta[n] + \delta[n - 1] + \delta[n - 2]$ and $y[n] = x[n] * h[n]$, where $h[n]$ is as defined above, plot $y[n]$ on the axis below.

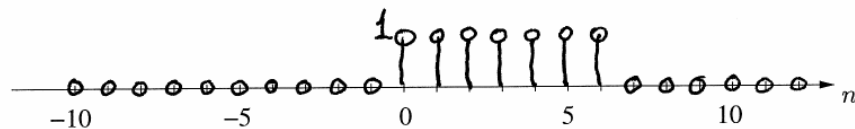


Exercise – 1

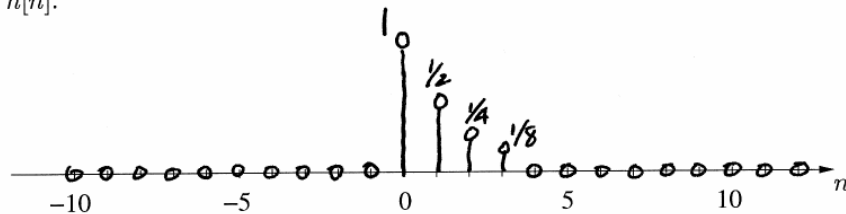


Let $x[n] = u[n] - u[n - 7]$ and $h[n] = \begin{cases} (\frac{1}{2})^n & 0 \leq n \leq 3 \\ 0 & \text{otherwise.} \end{cases}$

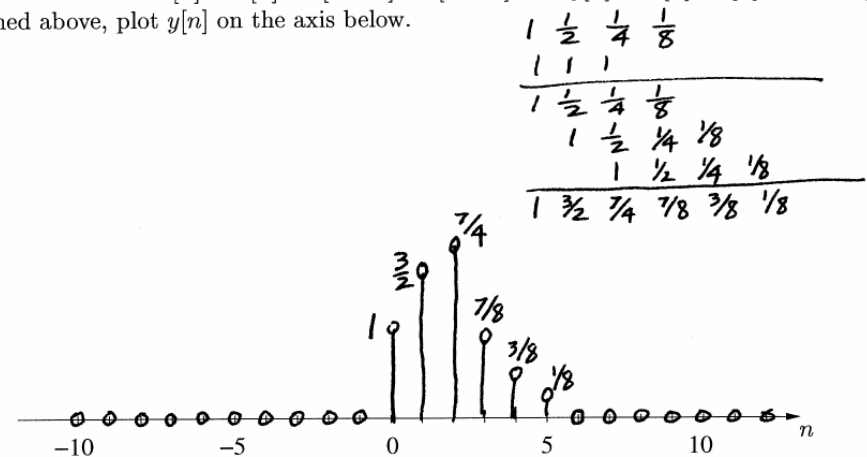
(a) Plot $x[n]$.



Plot $h[n]$.



(b) If we now assume $x[n] = \delta[n] + \delta[n - 1] + \delta[n - 2]$ and $y[n] = x[n] * h[n]$, where $h[n]$ is as defined above, plot $y[n]$ on the axis below.



Generalization of Discrete Time Systems

A linear, time-invariant system can be described by a difference equation having the following general form:

$$y(n) + a_1y(n-1) + \dots + a_Ny(n-N) = b_0x(n) + b_1x(n-1) + \dots + b_Mx(n-M)$$

$$y(n) = -a_1y(n-1) - \dots - a_Ny(n-N) + b_0x(n) + b_1x(n-1) + \dots + b_Mx(n-M)$$

$$y(n) = - \sum_{i=1}^N a_i y(n-i) + \sum_{j=0}^M b_j x(n-j)$$

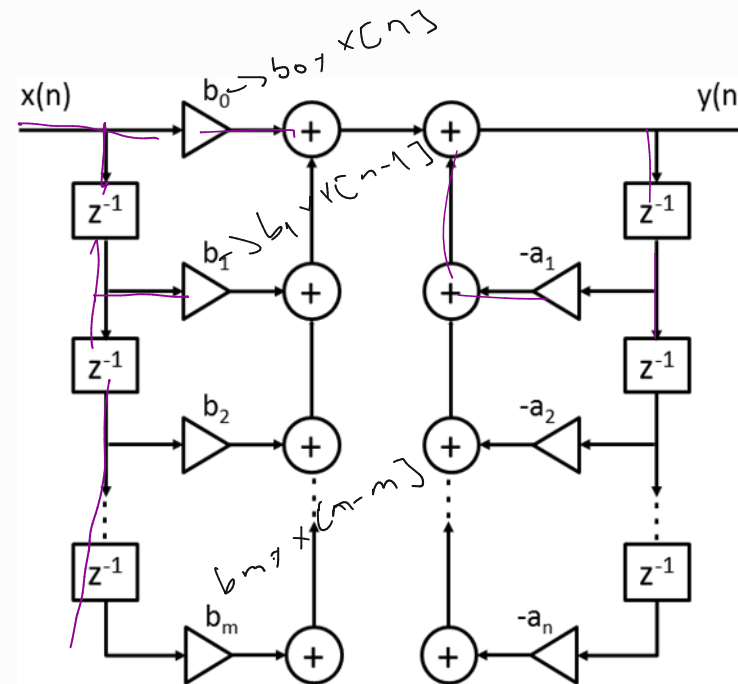
Linear Constant Coefficient Difference Equation

But "ayık" sistemi gösterir

$a=0$
 $b=5$
 $y[n] = 5x[n]$

Block Diagram Representation of LCCDE

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$



It is easy to implement the filters to hardware using block diagrams!

Example



Given the following difference equation:

$$y(n) = 0.25y(n-1) + x(n),$$

identify the nonzero system coefficients.

$$b_0 = 1$$

$$-a_1 = 0.25$$

Classification of Impulse Response $h[n]$

FIR – Finite Impulse Response:

- Number of impulses are limited.
- Always stable.

For example: $h[n] = \delta[n - 1] + 5\delta[n - 5]$

IIR – Infinite Impulse Response:

- Number of impulses are infinite.
- Sometimes these systems are not stable.

For example: $h[n] = u[n - 1] + 5u[n - 5]$

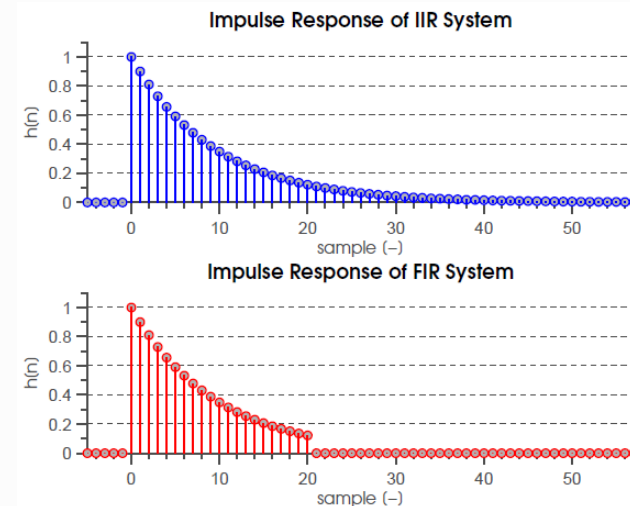
Finite impulse response (FIR):

$$y(n) = \sum_{k=0}^{M-1} h(k)x(n-k)$$

Infinite impulse response (IIR):

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$$

Another example:



If $N = 0$, the system has FIR

$$y(n) = -\sum_{i=1}^N a_i y(n-i) + \sum_{j=0}^M b_j x(n-j)$$

$$y(n) = \sum_{k=0}^M b_k x(n-k)$$

M th order FIR filter

Finite impulse response (FIR):

$$y(n) = \sum_{k=0}^M h(k) x(n-k)$$

Block Diagram Representation of FIR

- Direct Form

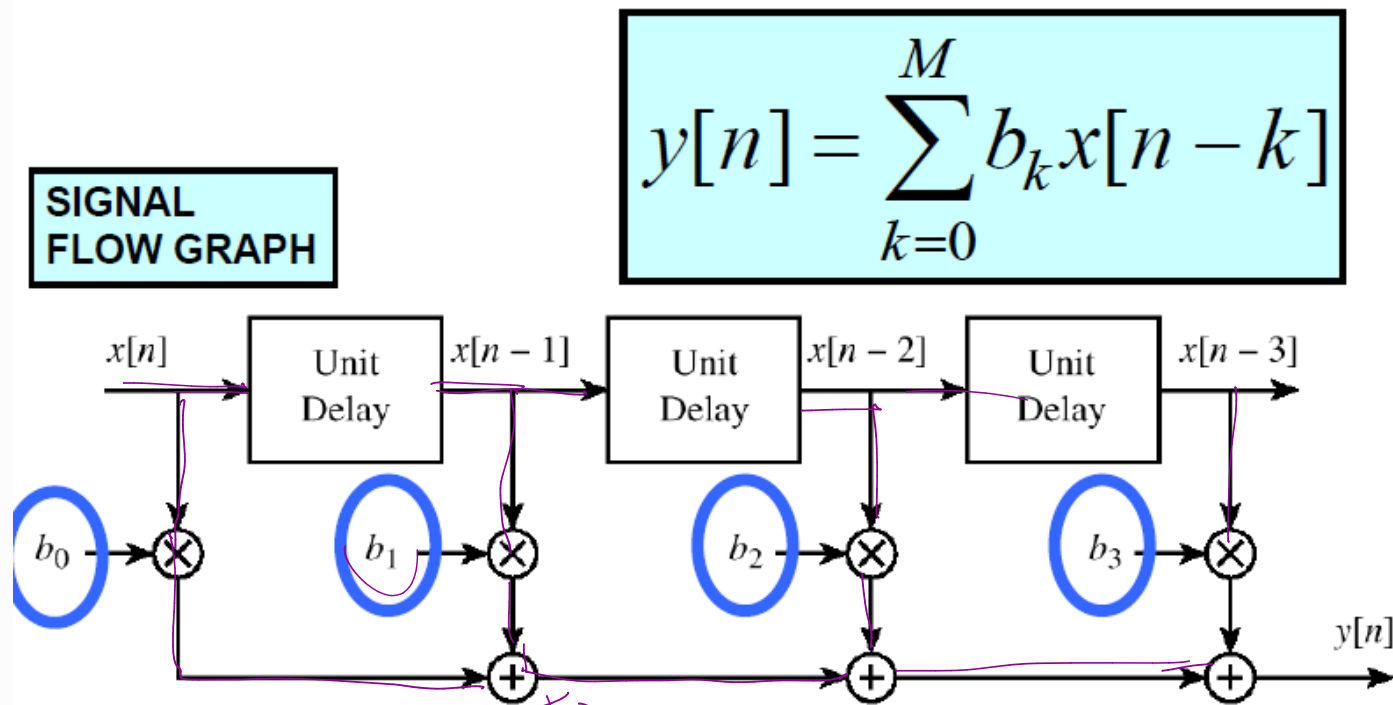
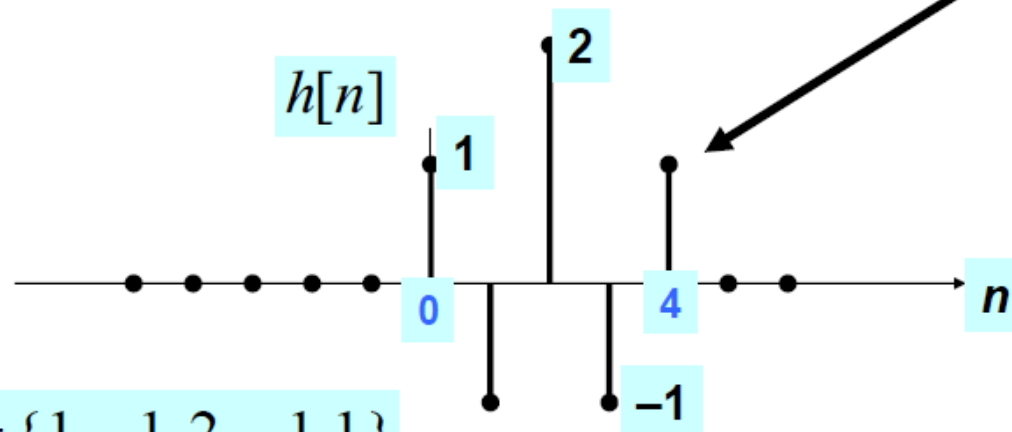


Figure 5.13 Block-diagram structure for the M th order FIR filter.

Math Formula of $h[n]$: FIR example

$$h[n] = \delta[n] - \delta[n-1] + 2\delta[n-2] - \delta[n-3] + \delta[n-4]$$



$$b_k = \{1, -1, 2, -1, 1\}$$

$$y[n] = \sum_{k=0}^M h[k]x[n-k]$$

Same as b_k

FINITE LIMITS

FINITE LIMITS

FIR Filter conv. - Table Method (Study it at home)

$$h[n] = \delta[n] - \delta[n-1] + 2\delta[n-2] - \delta[n-3] + \delta[n-4]$$

$$x[n] = u[n]$$

n	-1	0	1	2	3	4	5	6	7
$x[n]$	0	1	1	1	1	1	1	1	...
$h[n]$	0	1	-1	2	-1	1	0	0	0
$h[0]x[n]$	0	1	1	1	1	1	1	1	1
$h[1]x[n-1]$	0	0	-1	-1	-1	-1	-1	-1	-1
$h[2]x[n-2]$	0	0	0	2	2	2	2	2	2
$h[3]x[n-3]$	0	0	0	0	-1	-1	-1	-1	-1
$h[4]x[n-4]$	0	0	0	0	0	1	1	1	1
$y[n]$	0	1	0	2	1	2	2	2	...

3-point Average Filter

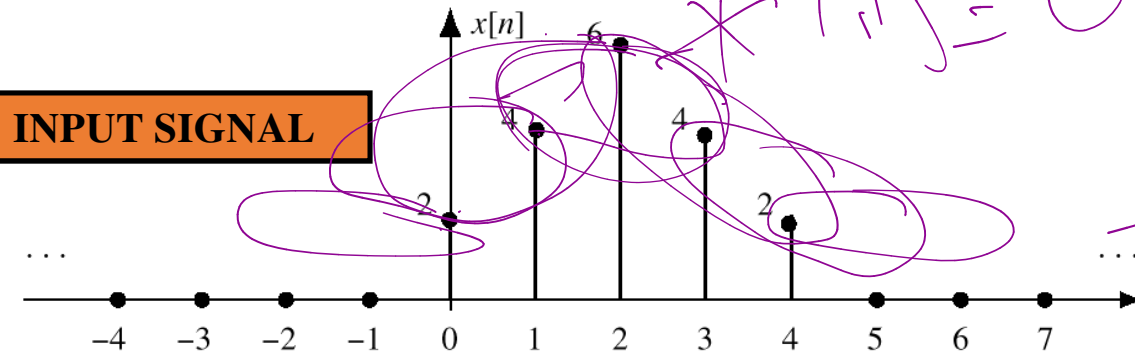


Figure 5.2 Finite-length input signal, $x[n]$.

$$y[n] = \frac{1}{3}(x[n] + x[n+1] + x[n+2])$$

$y[n] = \frac{1}{3}(x[n] + x[n+1] + x[n+2])$

non causal $\rightarrow$ real time

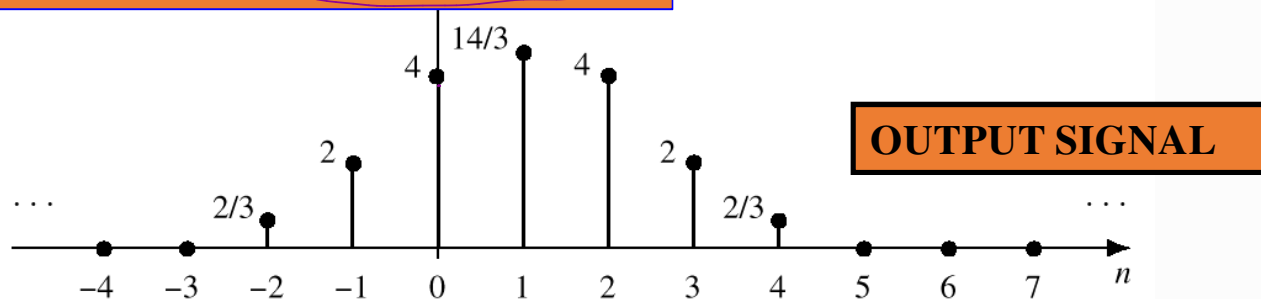


Figure 5.3 Output of running average, $y[n]$.

Is this system causal?

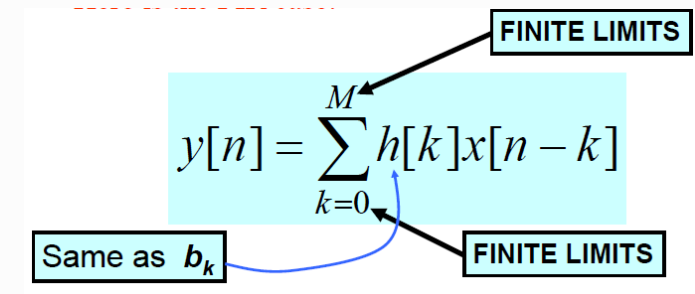
Do this system has FIR ?

$$y[n] = \sum_{k=0}^M h[k]x[n-k]$$

FINITE LIMITS (pointing to M)

Same as b_k (pointing to h[k])

FINITE LIMITS (pointing to 0)



4-point Average FIR Filter

$$y[n] = \frac{1}{4} (x[n] + x[n-1] + x[n-2] + x[n-3])$$

Find impulse response:

$$x[n] = \delta[n]$$

$$y[n] = \frac{1}{4} \delta[n] + \frac{1}{4} \delta[n-1] + \frac{1}{4} \delta[n-2] + \frac{1}{4} \delta[n-3]$$

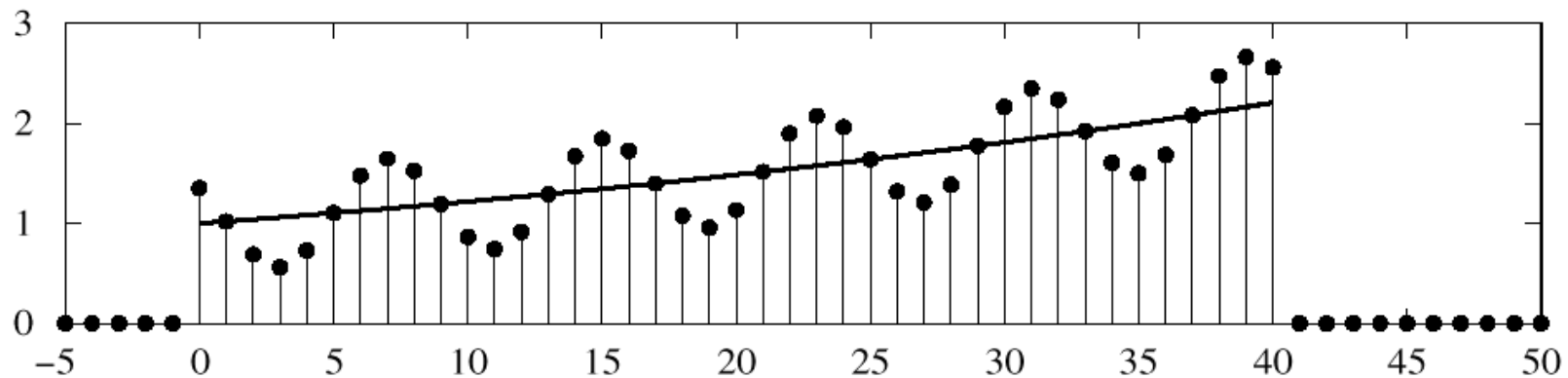
n	-3	-2	-1	0	1	2	3	4	5
$x[n]$	0	0	0	1	0	0	0	0	0
$y[n]$	0	0	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	0

$$h[n] = \{ \dots, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, \dots \}$$

↑
 $n=0$

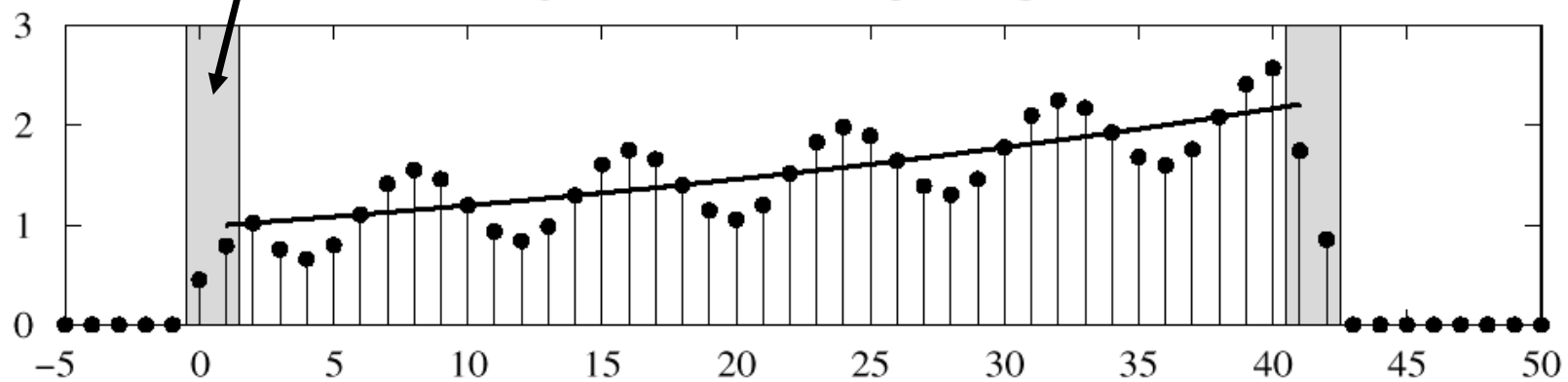
3-pt AVG EXAMPLE

Input : $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$



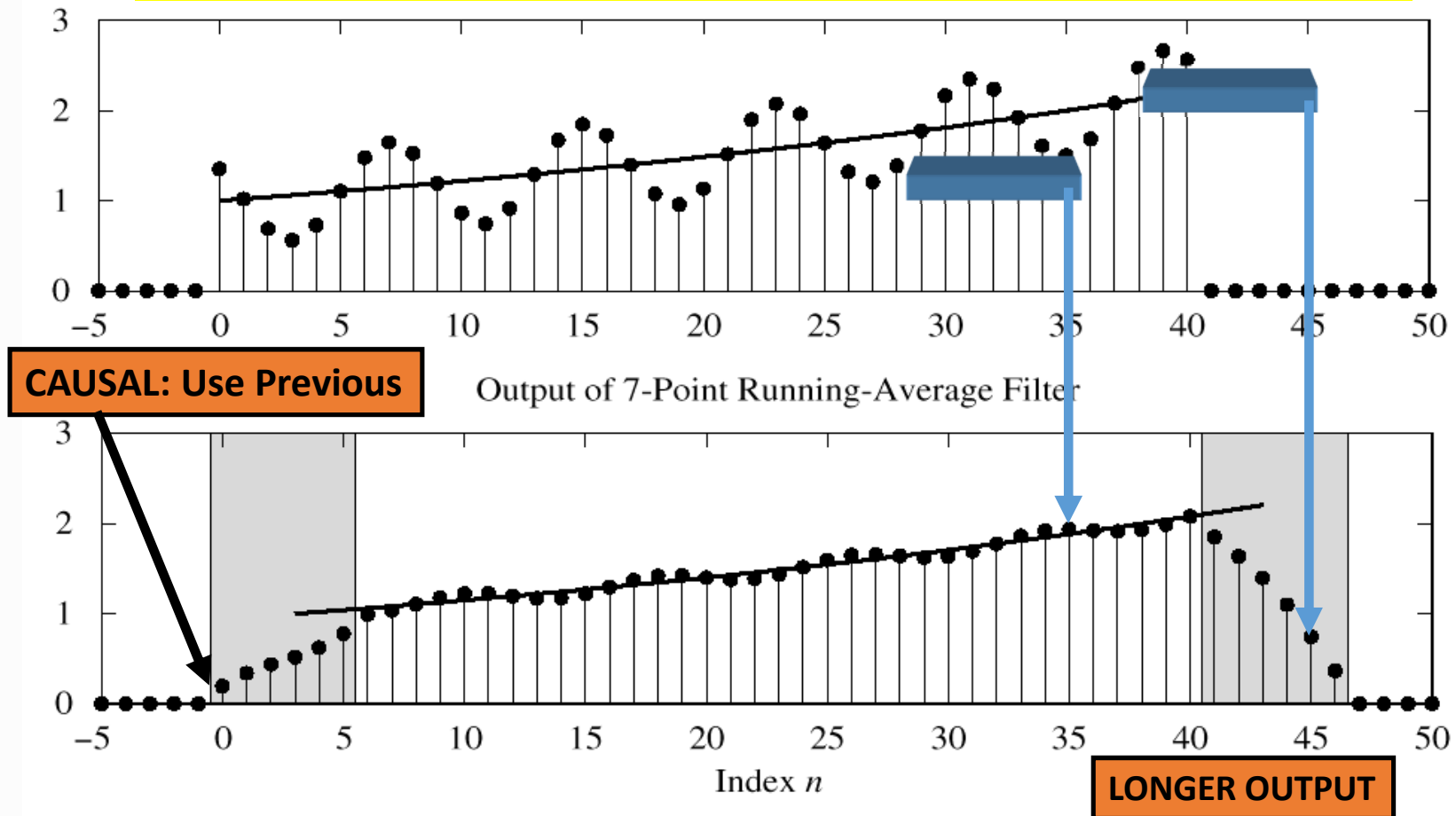
USE PAST VALUES

Output of 3-Point Running-Average Filter



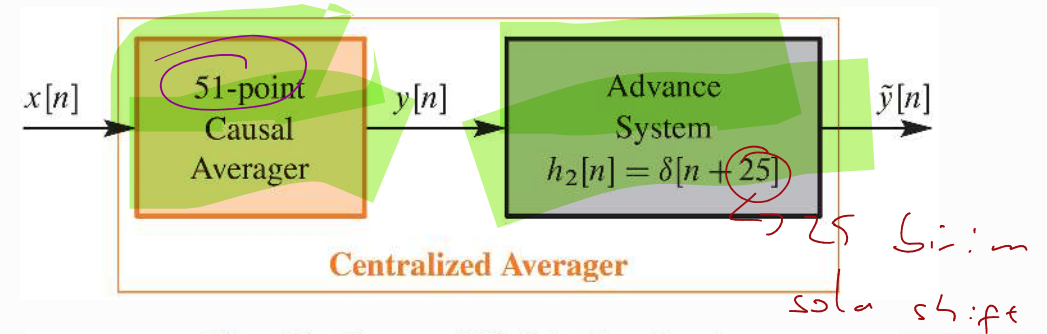
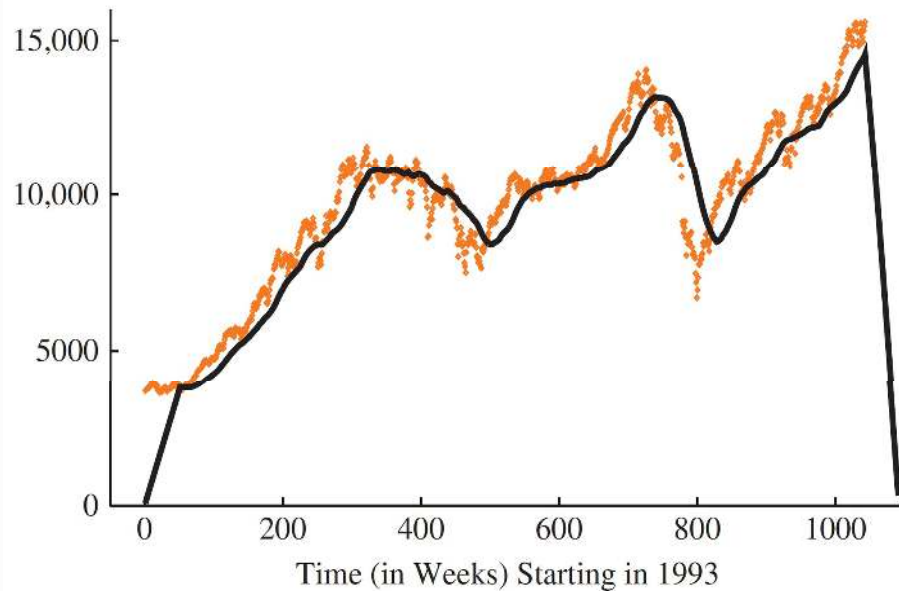
7-pt FIR EXAMPLE (AVG)

Input : $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$

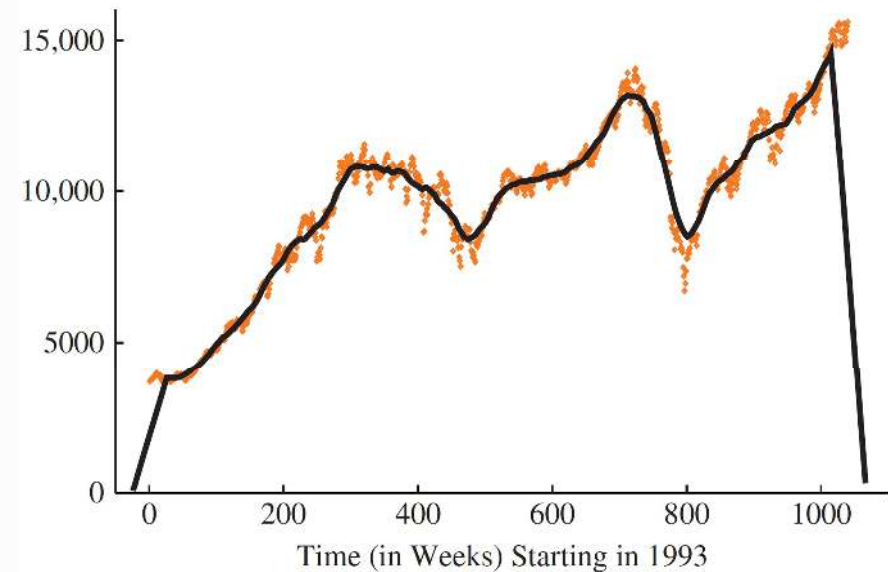


FILTER STOCK PRICES - CAUSAL VS ANTICAUSAL

Filtered by Causal 51-Point Running Averager



Filtered by Noncausal 51-Point Running Averager



Let's apply 17-pt Centralized Average filter to Noisy Audio

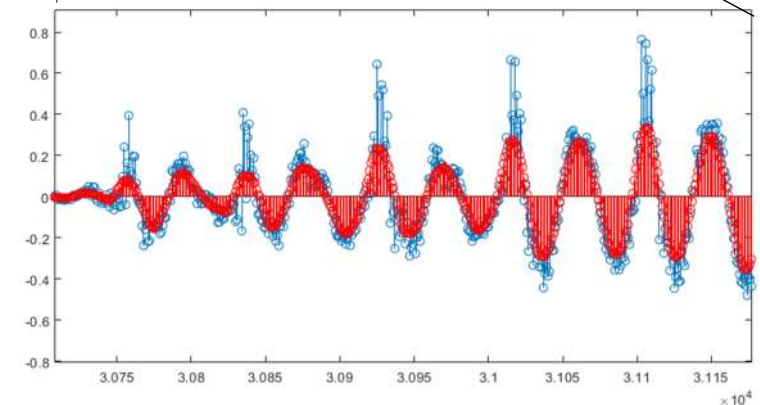
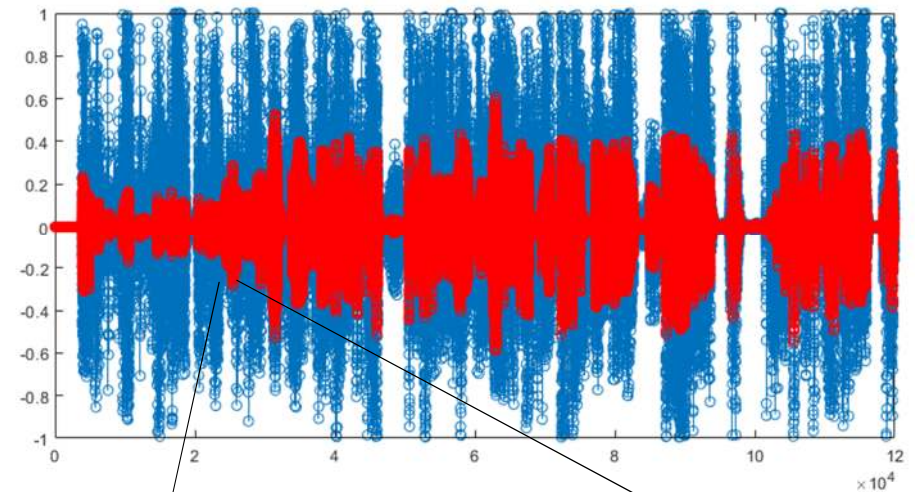
```
clc; clear all;

%% Load Sound
load ('piano2.mat');
x = x(1:16000);
soundsc(x,Fs);

%% Add noise
K = awgn(x,40);
soundsc(K,Fs);

%% Filter
N = 17;
h = 1/N*ones(1,N);

%% Apply Convolution
y = conv(K,h,'same');
soundsc(y,Fs);
%%
plot(x,'r'); hold on; plot(y,'b');
```



Apply Average Filter to An Image

```
clc; clear all;  
I = imread('eight.tif');  
I_noise =  
imnoise(I, 'gaussian', 0, 0.001);  
%%  
H = (1/9)*ones(3,3);  
ortalamaSonucu =  
conv2(I_noise, H, 'same');  
  
%%  
figure(1), imshow(I_noise, []);  
figure(2), imshow(ortalamaSonucu, []);
```

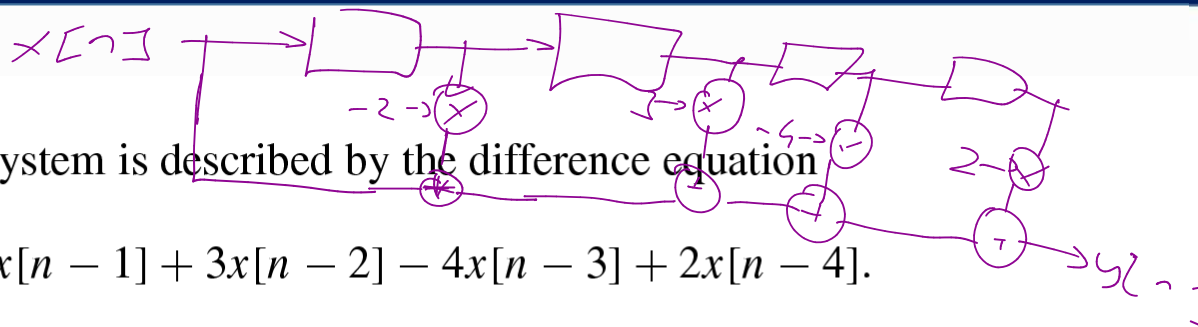
Exercise -1



PROBLEM:

A linear time-invariant discrete-time system is described by the difference equation

$$y[n] = x[n] - 2x[n-1] + 3x[n-2] - 4x[n-3] + 2x[n-4].$$



- Draw a block diagram that represents this system in terms of unit-delay elements, coefficient multipliers, and adders as in Figure 5.13 in the *SP First*.
- Determine the impulse response $h[n]$ for this system.

- Use convolution to determine the output due to the input

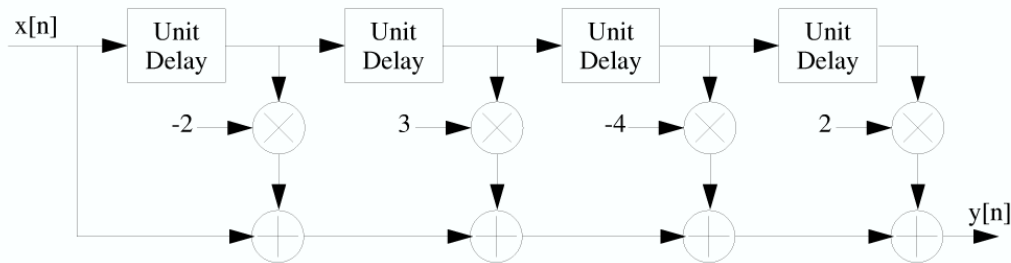
$$x[n] = \delta[n] - \delta[n-1] + \delta[n-2]$$

$$x[n] = \delta[n] - \delta[n-1] + \delta[n-2] = \begin{cases} 1 & n = 0, 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} y[0] &= 2 & y[2] &= 9 & y[4] &= 6 \\ y[1] &= -6 & y[3] &= -9 & y[5] &= -3 \\ y[6] &= 1 \end{aligned}$$

Plot the output sequence $y[n]$ for $-3 \leq n \leq 10$.

a) The block diagram for $y[n]$ is as follows.



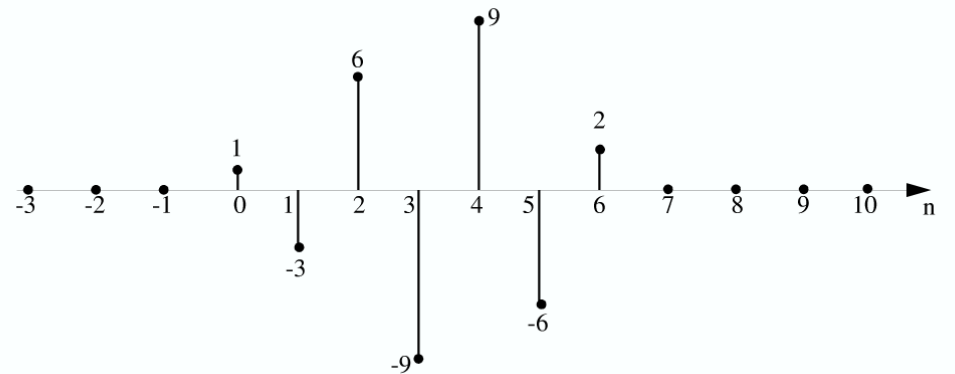
b) The impulse response for $y[n]$ can be found by using $x[n] = \delta[n]$ which results in

$$y[n] = h[n] = \delta[n] - 2\delta[n-1] + 3\delta[n-2] - 4\delta[n-3] + 2\delta[n-4]$$

c) $y[n]$ can be tabulated as follows.

n	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10
$h[n]$				1	-2	3	-4	2						
$x[n]$				1	-1	1								
$y[n]$				1	-2	3	-4	2						
					-1	2	-3	4	-2					
						1	-2	3	-4	2				
				1	-3	6	-9	9	-6	2				

Plotting $y[n]$ gives



Exercise -2



PROBLEM:

This problem is concerned with finding the output of an FIR filter for a given input signal. A linear time-invariant system is described by the difference equation

$$y[n] = \sum_{k=0}^4 (k+1)x[n-k]$$

Handwritten notes: $b_0 \rightarrow 1, b_1 \rightarrow 2, b_2 \rightarrow 3, b_3 \rightarrow 4, b_4 \rightarrow 5$. $h[n] = 1, 2, 3, 4, 5$. $x[n] = u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$. $n=5$. $-] + 2 \cdot x[n-1] -] + 4 \cdot x[n-3] -] + 5 \cdot x[n-4]$. $f[n] + 2 \cdot f[n-1] + 3 \cdot f[n-2] - - -$

The input to this system is *unit step* signal, denoted by $u[n]$, i.e., $x[n] = u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$

- Determine the filter coefficients $\{b_k\}$ of this FIR filter.
- Determine the impulse response, $h[n]$, for this FIR filter. The impulse response is a discrete-time signal, so make a (stem) plot of $h[n]$ versus n .
- Use convolution to compute $y[n]$, over the range $-5 \leq n \leq \infty$, when the input is $u[n]$. Make a plot of $y[n]$ vs. n . (Hint: you might find it useful to check your results with MATLAB's `conv()` function.)

Exercise -2



a) $y[n] = 1x[n] + 2x[n-1] + 3x[n-2] + 4x[n-3] + 5x[n-4]$

Filter coefficients $b_0=1 \quad b_1=2 \quad b_2=3 \quad b_3=4 \quad b_4=5$

($b_n=0$ for $n < 0$ and $n > 4$)

b) $h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3] + 5\delta[n-4]$

